

24/7 Autonomous Telegram Customer Assistant

RoboClicks Bot: Stateful Multi-Turn Conversational Agent with n8n & LangChain Memory

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Bot Identifier: @RoboClicksBot (Telegram Bot API)

Live Runtime: 24/7 Production Webhook • Zero Downtime Container

Memory Module: LangChain Window Buffer Memory (k=10 turns)

Intent Routing: Switch Node (Deterministic FAQs vs LLM Advisor)

Time Google Sheets Lead Pipeline

1. SYSTEM OVERVIEW & ARCHITECTURAL OBJECTIVE

Traditional chatbots fail either by being rigid button trees or expensive, slow hallucination-prone LLM calls. RoboClicks Bot implements a hybrid architecture: high-speed deterministic routing for standard queries (pricing packages at \$20, service specifications, /start onboarding), paired with a stateful LangChain agent that retains conversation history across 10 rolling interactions for nuanced technical consultations.

2. STEP-BY-STEP WORKFLOW & NODE ARCHITECTURE

1. Telegram Webhook Listener

Captures incoming chat updates, commands (/start), chat_id, and user handle with zero polling latency.

2. Input Sanitization & Session Check

Validates message structure, handles media guards, and initializes the conversation buffer session ID.

3. Intent Classification Switch Node

Routes immediate keyword/pricing inquiries to cached instant responses, saving token costs and reducing lag.

4. LangChain Agent + Window Buffer Memory

Injects rolling chat context into prompt template for personalized, coherent multi-turn technical advice.

5. CRM Lead Capture & Analytics Node

Asynchronously appends qualified user inquiries, contact timestamps, and intent scores into Google Sheets.

6. Telegram Bot Markdown Responder

Sends formatted Markdown responses featuring action prompts, inline keyboards, and clear next steps.

3. VALIDATION METRICS & PRODUCTION BENCHMARKS

- Response Latency (Deterministic FAQs): < 420 ms
- Response Latency (LangChain Agent + Memory): < 1.25 seconds
- Context Retention Test: 100% recall of user constraints stated 5+ turns earlier
- Lead Capture Accuracy: 100% logging reliability across simulated and live user test sessions